UNIT I – Foundations of Prompt Engineering

Experiment 1: Environment & Connectivity

AIM:

To install the required Python libraries, securely configure the API key, establish connectivity with the Large Language Model (LLM), and verify successful model access by executing a simple prompt ("Hello, World").

DESCRIPTION:

Large Language Models (LLMs) require proper environment setup and API configuration before they can be used for prompt engineering tasks. This experiment introduces the basic workflow for establishing communication between a Python application and an LLM using an API.

The experiment focuses on setting up the Python environment by installing the required libraries, configuring the API key securely using a .env file, and establishing connectivity with the Large Language Model (LLM). After creating the client and loading the API credentials, a simple **"Hello, World!"** prompt is executed to verify successful communication with the model. The successful response confirms that the environment and API are correctly configured, providing the foundation for all subsequent prompt engineering experiments.

PROGRAM:

```
# Step 1: Import the Google GenAI library
from google import genai

# Step 2: Create the Gemini client using your API key
client = genai.Client(
    api_key="YOUR_GEMINI_API_KEY"
)

# Step 3: Display a heading
print("GOOGLE GEMINI ENVIRONMENT & CONNECTIVITY TEST\n")

# Step 4: Send a prompt to the Gemini model
response = client.models.generate_content(
    model="gemini-3.6-flash",
    contents="Print only: Hello World"
)

# Step 5: Display the generated response
print("Response from Gemini:\n")
print(response.text)
```

OUTPUT:

GOOGLE GEMINI ENVIRONMENT & CONNECTIVITY TEST

Response from Gemini:

Hello World